Hao Luan

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Education

National University of Singapore Singapore, SG
Doctor of Philosophy (Ph.D.) in Computer Science *Aug. 2023 –*

- Advisor: Prof. Zhiyong Huang

Harbin Institute of Technology, Shenzhen Shenzhen, CHN
Bachelor of Engineering (B.Eng.), Automation *Sep. 2017 – Jun. 2021*

- Overall GPA: 90.1/100, 3.8/4.0

Remarks:

- Admitted with full fellowship to the *Master of Applied Science* (research-based) program in 2022 and the *Direct Entry PhD* program in 2023 at the Edward S. Rogers Sr. Department of Electrical and Computer Engineering, University of Toronto, Canada. Not matriculated due to severe visa delays.

Experiences

Huang's Group – National University of Singapore Jun. 2026 –
Graduate Researcher Singapore, SG
Advisor: Prof. Zhiyong Huang *Comp. Sci.*

- Researching on constrained diffusion models for generative decision-making.

AGORAi Lab – NUS Jul. 2024 – May 2026
Graduate Researcher Singapore, SG
Advisors: Prof. Chun Kai Ling, Prof. See-Kiong Ng *Comp. Sci.*

- Research on inference-time approaches for constrained diffusion generation.

Collaborative, Learning, and Adaptive Robots Lab – NUS Aug. 2023 – May 2024
Graduate Researcher Singapore, SG
Advisor: Prof. Harold Soh *Comp. Sci.*

- Proposed policy diffusion-based algorithms to satisfy *spatial-temporal symbolic* constraints for safe and customized robot planning.

Robotics Perception & Intelligence Lab – SUSTech & CUHK Aug. 2021 – Jul. 2022
Research and Teaching Assistant Shenzhen, CHN
Advisors: Prof. Max Q.-H. Meng, Prof. Jiankun Wang *Electronic & Electrical Eng.*

- Developed planning and control modules for a robotic solution to autonomous trolley collection and collaborative transportation at airports.

Peng Bo Technology (Shenzhen) Co. Ltd. Mar. 2021 – Apr. 2021
Software Development Intern Shenzhen, CHN
Supervisor: Dr. Shixin Mao

- Developed low-level drivers for the company's cleaning robot products.

Multi-Agent Systems Lab – Harbin Inst. Tech. Shenzhen Oct. 2019 – Jun. 2021
Undergraduate Research Assistant Shenzhen, CHN
Advisor: Prof. Jie Mei *Automation*

- Conducted theoretical research on multi-agent systems consensus control over directed topologies.

Robot. Automat. Percep. & Decis. Lab – Sun Yat-sen Univ. Nov. 2015 – May 2016
High School Research Intern Guangzhou, CHN
Advisor: Prof. Hui Cheng *Comp. Sci. & Eng.*

- Optimized and implemented a centralized offline task-allocation algorithm for multi-robot systems.

Publications & Preprints

* indicates co-first authorship.

Preprints

- **H. Luan**, S.-K. Ng, and C. K. Ling, "Inference-time attribute distribution alignment for unconditional diffusion," *In Submission, arXiv: 2605.07456*, 2026. [`arXiv`]
- Z. Zheng, Z. Meng, **H. Luan**, W. Liu, W. S. Lee, "One token per multimodal evidence: Latent memory for resource-constrained QA," *In Submission, arXiv: 2606.10572*, 2026. [`arXiv`] [`Code`]

Conference

[C1] **H. Luan***, Y. X. Goh*, S.-K. Ng, and C. K. Ling, "Projected coupled diffusion for test-time constrained joint generation," *The Fourteenth International Conference on Learning Representations*, 2026, url: `https://openreview.net/forum?id=1FEm5JLpvg`. [`arXiv`] [`Code`]

[C2] Z. Feng, **H. Luan**, K. Y. Ma, and H. Soh, "Diffusion meets options: Hierarchical generative skill composition for temporally-extended tasks," *2025 International Conference on Robotics and Automation (ICRA)*, 2025, pp. 10854–10860, doi: `10.1109/ICRA55743.2025.11127641`. [`arXiv`] [`Page`]

[C3] J. Zhao, H. Ye, Y. Zhan, **H. Luan**, H. Zhang, "Human orientation estimation under partial observation," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 11544–11551, doi: `10.1109/IROS58592.2024.10802390`. [`arXiv`]

[C4] B. Xia*, **H. Luan***, Z. Zhao*, X. Gao, P. Xie, A. Xiao, J. Wang, and M. Q.-H. Meng, "Collaborative trolley transportation system with autonomous nonholonomic robots," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 8046–8053, doi: `10.1109/IROS55552.2023.10341508`. [`arXiv`] [`PDF`] [`Video`]

[C5] A. Xiao*, **H. Luan***, Z. Zhao*, Y. Hong, J. Zhao, W. Chen, J. Wang, and M. Q.-H. Meng, "Robotic autonomous trolley collection with progressive perception and nonlinear model predictive control," *2022 International Conference on Robotics and Automation (ICRA)*, 2022, pp. 4480–4486, doi: `10.1109/ICRA46639.2022.9812455`. [`Page`] [`arXiv`] [`PDF`] [`Video`]

Journal

[J1] Z. Feng*, **H. Luan***, P. Goyal, and H. Soh, "LTLDoG: Satisfying temporally-extended symbolic constraints for safe diffusion-based planning," *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 10, pp. 8571–8578, 2024, doi: `10.1109/LRA.2024.3443501`. [`arXiv`] [`PDF`] [`Page`] [`Code`]

[J2] X. Gao, **H. Luan**, B. Xia, Z. Zhao, J. Wang, and M. Q.-H. Meng, "A divide-and-conquer control strategy with decentralized control barrier function for luggage trolley transportation by collaborative robots," *Robotica*, vol. 41, no. 11, pp. 3333–3348, 2023, doi: `10.1017/S0263574723001005`. [`Video`]

[J3] **H. Luan**, J. Mei, A.-G. Wu, and G. Ma, "Distributed constrained consensus of multi-agent systems with uncertainties and disturbances under switching directed graphs," *IEEE Transactions on Control of Network Systems*, vol. 11, no. 1, pp. 161-172, 2024, doi: `10.1109/TCNS.2023.3272848`. [`Page`] [`PDF`]

Workshop

[W1] **H. Luan**, S.-K. Ng, and C. K. Ling, "DDPS: Discrete diffusion posterior sampling for layered graphs," *ICLR 2025 Workshop on Frontiers in Probabilistic Inference: Learning meets Sampling*, 2025, url: `https://openreview.net/forum?id=DBdkU0Ikzy`. [`arXiv`]

Selected Awards & Fellowships

NUS Research Scholarship, *National University of Singapore*	2023 – 2027
Edward S. Rogers Sr. Graduate Scholarship, *University of Toronto* (Declined)	2023 – 2028
Outstanding Bachelor's Thesis (top 2%), *HIT Shenzhen*	2021
Mathematical Contest In Modeling (MCM) Honorable Mention	2020
Undergraduate Academic Merit Scholarship, *HIT Shenzhen*	2018, 2019, 2020
National Olympiad in Informatics in Provinces (NOIP) Third Prize, *China Computer Federation*	2016
American Mathematics Contest (AMC) 12 Honor Roll (invited to AIME)	2016

Academic Services

Conference Reviewing

- The International Conference on Machine Learning (ICML), Gold Reviewer
- The Conference on Neural Information Processing Systems (NeurIPS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Journal Reviewing

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Automation Science and Engineering (T-ASE)

Skills

Languages: Mandarin Chinese (native), Cantonese (native), English (full professional proficiency)

Programming: Python, C/C++, Julia, Pascal

Tools: CasADi, Claude Code, Codex, Git, MATLAB/Simulink, PyTorch, ROS, LaTeX